

Figure 1.25

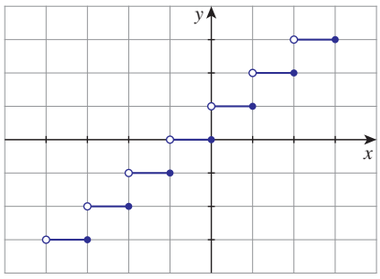


Figure 1.26

One may note that the relations among the names of these functions, the symbols denoting the functions are the commonly used words ceiling and floor of a room. Their graphs are given in Figures 1.25 and 1.26.

1.6.3 Types of Functions

Though functions can be classified into various types according to the need, we are going to concentrate on two basic types: *one-to-one functions and onto functions*.

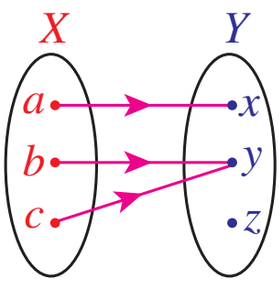


Figure 1.27

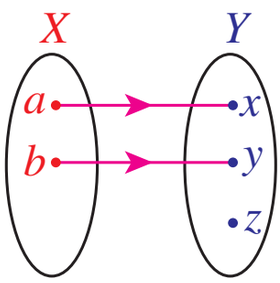


Figure 1.28

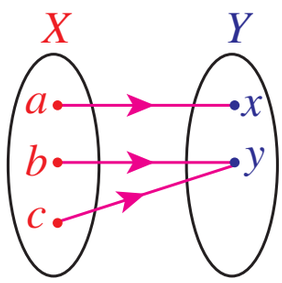


Figure 1.29

Let us look at the two simple functions given in Figure 1.27 and Figure 1.28. In the first function two elements of the domain, b and c, are mapped into the same element y, whereas it is not the case in the Figure 1.28. Functions like the second one are examples of one-to-one functions.

Let us look at the two functions given in Figures 1.28 and 1.29. In Figure 1.28 the element z in the co-domain has no pre-image, whereas it is not the case in Figure 1.29. Functions like in Figure 1.29 are examples of onto functions. Now we define one-to-one and onto functions.

Definition 1.7

A function $f : A \to B$ is said to be ***one-to-one*** if $x, y \in A, x \neq y \Rightarrow f(x) \neq f(y)$ [or equivalently $f(x) = f(y) \Rightarrow x = y$]. A function $f : A \to B$ is said to be ***onto***, if for each $b \in B$ there exists at least one element $a \in A$ such that $f(a) = b$. That is, the range of f is B.

Another name for one-to-one function is ***injective function***; onto function is ***surjective function***. A function $f : A \to B$ is said to be ***bijective*** if it is both one-to-one and onto.

To prove a function $f : A \to B$ to be one-to-one, it is enough to prove any one of the following:

if $x \neq y$, then $f(x) \neq f(y)$, or equivalently if $f(x) = f(y)$, then $x = y$.